

Ahmed Ikram | Software Engineer

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Final-year BSc Computer Science (Data Science & AI), University of Dundee (predicted First, June 2027). Ship full-lifecycle software - design to deploy to real-user evaluation, fixing the issues users surfaced before deployment. Learned Terraform and cloud networking from zero inside an industry project where I was Team Lead for a 7-person Agile team. Work in standard PR and code-review conventions in university group project teams and stayed composed with difficult customers from front-of-house retail. Mentored junior pupils one-to-one as a classroom assistant. Seeking a graduate Software Engineering role from June 2027.

EDUCATION

University of Dundee - BSc (Hons) Computer Science (Data Science and AI) *Sep 2023 - Present*

- On track for a First-Class Honours degree. Expected Graduation: June 2027

High School of Dundee *Sep 2018 - June 2023*

- Highers: A Mathematics, A Computing Science, A English, B Chemistry
- Computing Science classroom assistant - mentored junior pupils one-to-one during classwork.

TECHNICAL SKILLS

Languages: Python · TypeScript · JavaScript · SQL · T-SQL · Rust / PyO3 · Bash

Backend, APIs, Databases: FastAPI · Flask · asyncio · GraphQL (Strawberry) · Pydantic · SQLAlchemy · Apache Kafka · Model Context Protocol (MCP) · PostgreSQL · SQLite · Redis · Alembic

Frontend: React · React Native · Expo · Tailwind CSS · Vite · Formik · GraphQL Codegen · graphql-ws · Socket.IO

Cloud & DevOps: AWS (ECS Fargate, EC2, S3, RDS, ElastiCache, WAF, IAM, CloudFront, Secrets Manager, DynamoDB) · GCP (GKE, Cloud SQL, Cloud Monitoring) · Kubernetes (Kustomize, HPA, PDB, NetworkPolicy) · Azure (Databricks, Azure SQL, ADLS) · Terraform · Docker · Linux · GitHub Actions

Observability & Testing: Grafana · Prometheus · OpenTelemetry (Jaeger) · StatsD · Pytest · Jest · Playwright · Cypress · k6

PROJECTS

DevSync Project Management Platform with GitHub Integration

React · Tailwind · Flask · PostgreSQL · Socket.IO · GitHub OAuth · GCP · GKE · Terraform · GitHub Actions CI/CD

- Realtime boards (tasks, live comments, presence) on JWT Socket.IO rooms that re-verify project membership per-event and degrade gracefully instead of crashing; Redis message queue fans out across replicas with cross-pod receipt tests.
- Zero-long-lived-credentials: GCP workload identity, digest-pinned, Trivy-gated images signed with cosign & SLSA L3 provenance & SBOM attestation, default-deny network policy, Terraform-managed cluster.
- Deployed to GKE: HPA on CPU, PDBs, private-IP Cloud SQL, ephemeral per-PR environments; every push to main redeploys automatically and is re-verified against the live URL (health, OAuth config, k6) before the pipeline completes.
- GitHub OAuth integration: issue/PR linking and comment write-back with encrypted token storage; 3-tier RBAC enforced server-side.
- 1,505 tests and CodeQL static analysis gate every merge, p95 232ms vs a 500ms ceiling at 250/250 passing checks; a google managed Prometheus dashboard flags backend downtime.

ATM Log Aggregation, Anomaly Detection & Diagnostics (NCR Atleos)

FastAPI · React · PostgreSQL · Kafka · Redis · Docker · Terraform · AWS · CI/CD

- Architected a FastAPI REST API (30 endpoints, 6 routers, JWT & Redis-blacklist auth) with health-check-cascading startup across Kafka, Redis, and PostgreSQL.
- Built the Kafka (KRaft) event pipeline: at-least-once delivery, hybrid Redis/LRU deduplication, Redis Stream DLQ, sustaining ~100 msg/s over 2.5M events; W3C trace context via Kafka headers end-to-end, with OpenTelemetry Collector exporting traces to Jaeger.
- Applied 8 Redis patterns (rate-limit, dedup, lock, pub/sub, cache, JWT blacklist, DLQ, counters) off one connection pool; React frontend ships as a ~25MB nginx-alpine image measured at build time.
- Provisioned Terraform across a multi-AZ VPC (ECS Fargate behind ALB, EC2-hosted Kafka & Redis in a private subnet); a 3-pipeline CI/CD runs 1,887 tests (1,307 pytest, 495 Vitest, 10 Playwright E2E, 75 Terraform assertions) and Checkov compliance on every PR, rolling out zero-downtime with zero long-lived credentials. State locked in DynamoDB.

StockLens FinTech Spending & Investment Mobile App

React Native · Expo · FastAPI · PostgreSQL · Redis · Rust · GraphQL · MCP · Terraform · Docker · AWS · CI/CD

- Engineered an async FastAPI backend: 70+ REST endpoints with JWT auth (bcrypt), biometric unlock, and a stateless MCP server exposing 16 tools over streamable HTTP with OAuth 2.1 PKCE and JWT/JWKS validation.
- Replaced a pandas bottleneck with a native Rust extension (PyO3/Maturin, 12 functions) for technical-indicator computation, cross-built for ARM64 via QEMU in CI and shared by both the LSTM training pipeline and live API.
- Added a GraphQL read façade over the same service layer as REST: Strawberry resolvers with batched preloads eliminate N+1 queries, backed by a shared 60-second Redis quote cache; an authenticated graphql-ws subscription replaces client polling, and agent responses stream over an SSE endpoint via a custom fetch+ReadableStream parser.
- Deployed via Terraform (14 modules, 164 resources): three-tier network isolation (WAF-protected ALB → ECS Fargate ARM64 → private RDS/Redis) for a 20-30% compute-cost cut over x86; multi-stage images are 450MB vs 1.2GB naive.
- 1,584 backend and 844 frontend assertions - plus CodeQL static analysis on the frontend - gate a 9-job CI and an 8-job, 5-stage OIDC-authenticated CD pipeline.

W3C Web Logs ETL Pipeline

Airflow · Databricks · dbt · PySpark · Azure SQL · Terraform · Power BI · Grafana · Prometheus · Marquez · OpenLineage

- Architected a Databricks DLT medallion in PySpark (Bronze → Silver in DLT; gold via dbt) on serverless distributed compute; 153,380 rows ingested with 3 Silver rows dropped through 10 DLT quality gates (7 bronze & 3 Silver).
- 16 dbt models compile against PostgreSQL and T-SQL from one codebase via 18 compatibility macros; 121 dbt tests gate every model, catching 213 duplicate rows through a 12-component deduplication key.
- Integrated OpenLineage with a self-hosted Marquez collector across 2 Airflow DAGs - task-run events, dbt model/column lineage, Azure SQL dataset edges, and custom data-quality facets in one cross-engine lineage graph.
- Deployed 30+ Azure resources through Terraform and OIDC federation; 630 tests, a freshness probe, 3 Grafana dashboards (23 panels), 8 Prometheus rules and 2 Azure Monitor alerts validate pipeline health and data quality.

CERTIFICATES

<u>AWS Academy Graduate - Machine Learning Foundations</u>	<i>Feb 2025</i>
<u>Foundations of Azure AI: Concepts, Capabilities, and Implementation</u>	<i>Oct 2024</i>
<u>AWS Academy Graduate - Cloud Foundations</u>	<i>Mar 2024</i>

EXPERIENCE

Beaully Mini Market , Dundee, Scotland	
<i>Retail Assistant</i>	<i>Sep 2023 - Present</i>
Zulus Peri Peri Grill , Dundee, Scotland	
<i>Front of House Worker</i>	<i>June 2021 - Sep 2023</i>